Translations

1 Consider the parabola $y = x^2 - 3$.

a Complete the table of values:

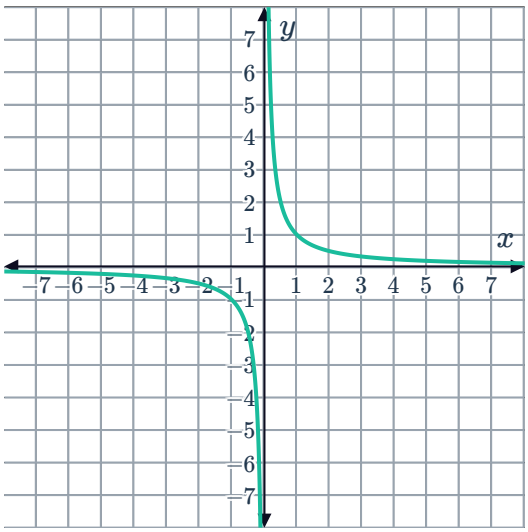
x	-2	-1	0	1	2
y					

b Sketch the graph of $y = x^2 - 3$.

c What is the y-intercept of the graph?

d When adding a constant to the equation $y = x^2$, describe the type of transformation that occurs on its graph.

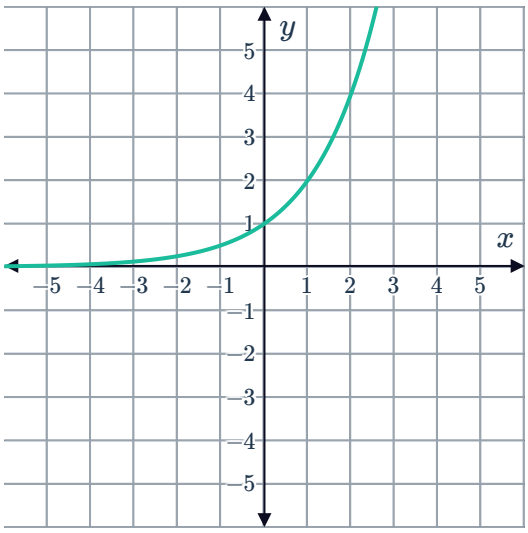
2 Consider the graph of $y = \dfrac{1}{x}$:



a What translation is required to shift the graph of $y = \dfrac{1}{x}$ to get the graph of $y = \dfrac{1}{x} + 4$.

b Hence, sketch $y = \dfrac{1}{x} + 4$.

3 Consider the graph of $y = 2^x$:

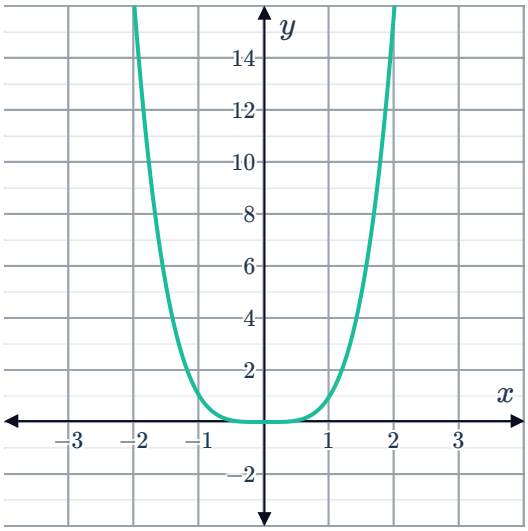


a What translation is required to shift the graph of $y = 2^x$ to get the graph of $y = 2^x - 5$.

b Hence, sketch $y = 2^x - 5$.

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4 A graph of $y = x^4$ is shown. Sketch the curve after it has undergone a transformation resulting in the function $y = x^4 - 2$.



5

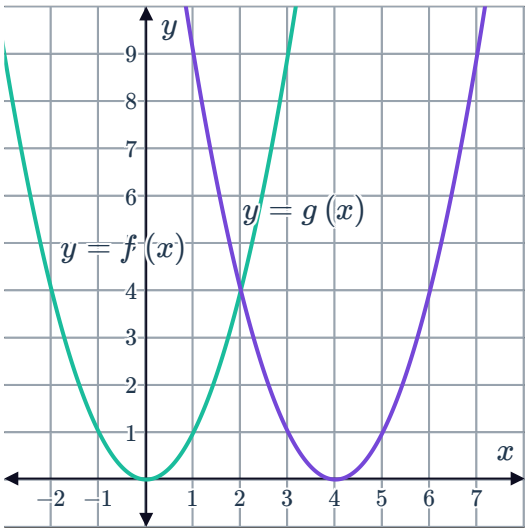
a How should you translate the graph of $y = f(x)$ to get the graph of $y = f(x) + 4$?

b How should you translate the graph of $y = g(x)$ to get the graph of $y = g(x + 6)$?

6 The functions $f(x)$ and $g(x) = f(x + k)$ have been graphed:

a Determine the value of k.

b Describe the transformation that occured.



7 Describe the shift required to transform the graph of $y = a^x$ to get the graph of $y = a^{(x+5)}$.

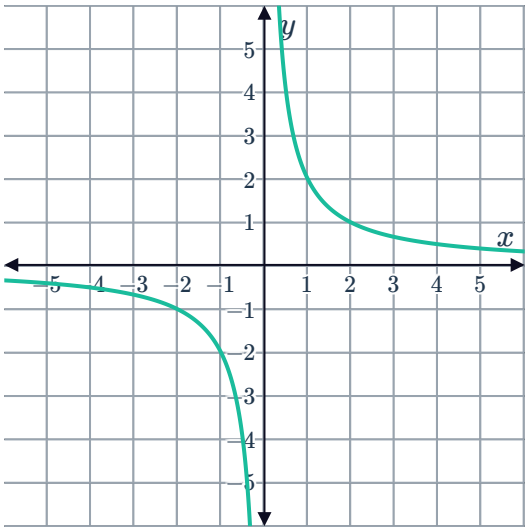
8 For each of the following equations, determine what the new equation is when their graphs are moved as described:

a The graph of $y = x^3$ is moved to the right by 10 units.

b The graph of $y = 2^x$ is moved down by 9 units.

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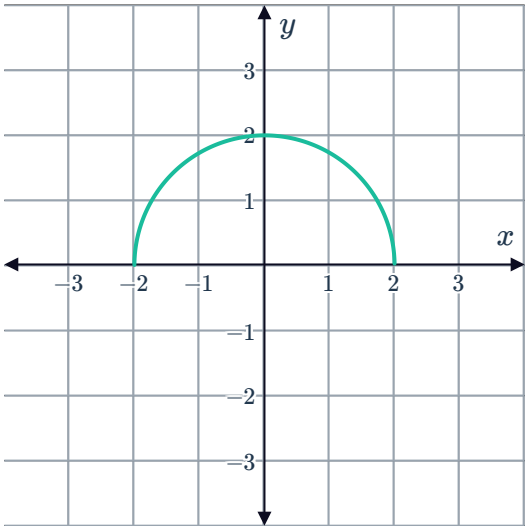
9 Consider the graph of the hyperbola $y = \frac{2}{x}$:



a What would be the new equation if the graph was shifted upwards by 2 units?

b What would be the new equation if the graph was shifted to the right by 9 units?

10 Consider the graph of $y = \sqrt{4 - x^2}$:

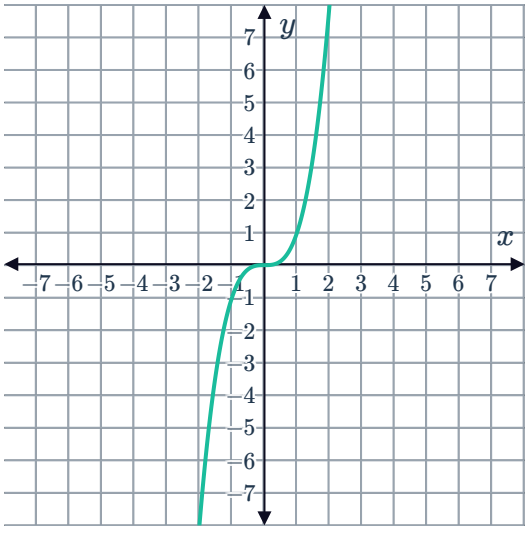


a What would be the new equation if the graph was translated downwards by 7 units?

b What would be the new equation if the graph was translated to the left by 3 units?

11 If the graph of $y = x^4$ is moved to the right by 8 units and up by 6 units, what is its new equation?

12 Consider the graph of $y = x^3$:



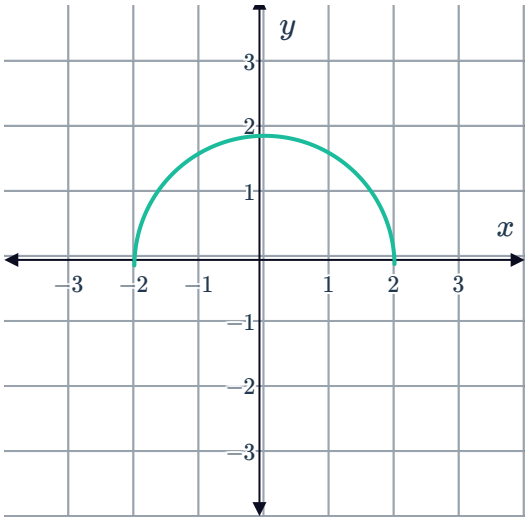
a Describe the required translations to shift the graph of $y = x^3$ to get the graph of $y = (x + 2)^3 + 4$.

b Hence, sketch $y = (x + 2)^3 + 4$.

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a Describe the required translations to shift the graph of $y = \sqrt{4 - x^2}$ to get the graph of $y = \sqrt{4 - x^2} + 2$.

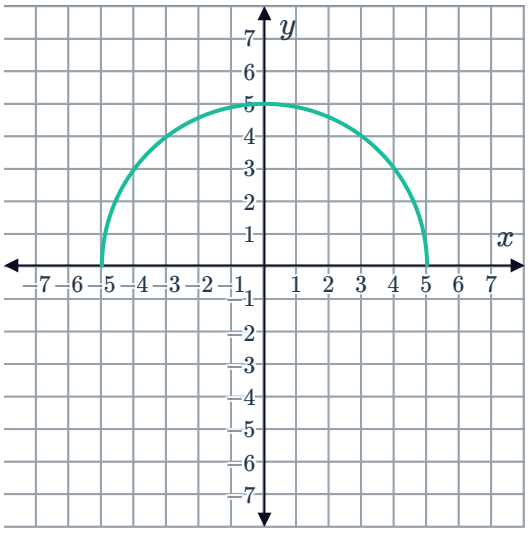
b Hence, sketch $y = \sqrt{4 - x^2} + 2$.



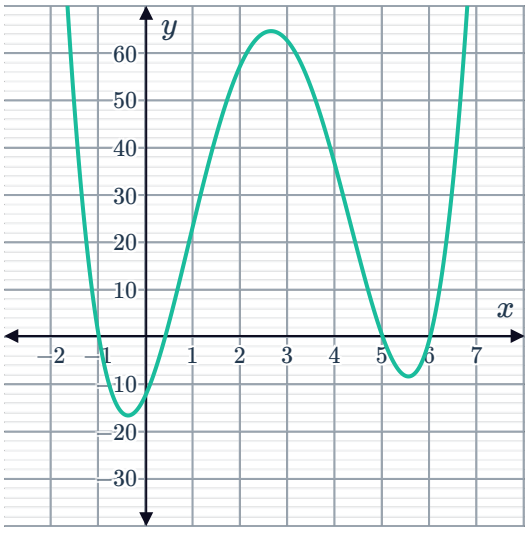
14 Consider the graph of $y = \sqrt{25 - x^2}$:

a Describe the required translations to shift the graph of $y = \sqrt{25 - x^2}$ to get the graph of $y = \sqrt{25 - (x + 4)^2} - 2$.

b Hence, sketch $y = \sqrt{25 - (x + 4)^2} - 2$.



15 The graph of $y = P(x)$ is shown. Sketch the graph of $y = P(x) - 20$.



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Reflections

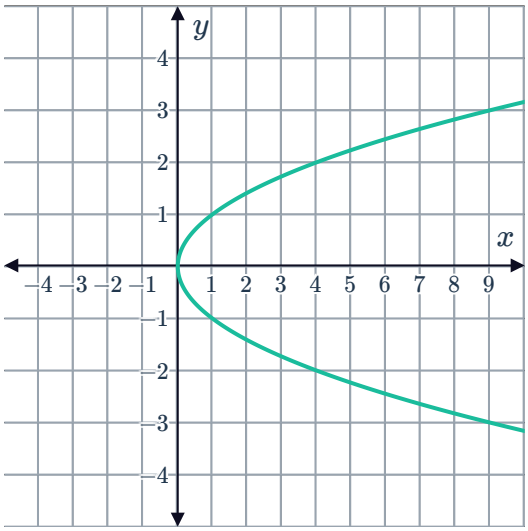
16 Consider the point $P\,(8, -3)$. Sketch the point that is symmetric to $P\,(8, -3)$ with respect to:

a The x-axis **b** The y-axis **c** The origin

17 State whether the following functions have symmetry. If so, state the line or point of symmetry.

a $y = x^2 + 1$ **b** $y = x + 5$

18 Is the graph shown symmetric with respect to the x-axis, the y-axis, or the origin?



19 Does the function $y = 4x^3$ have reflective symmetry?

20 For each of the following functions:

i Find $f(-x)$.

ii State whether the functions is symmetrical along the x-axis, the y-axis or neither.

a $f(x) = 2x^3 - 2x$ **b** $f(x) = 0.9x^6 - 7x^2 + 4$

c $f(x) = x^3 + x - 3$ **d** $f(x) = \dfrac{1}{6x^3}$

21 Consider the point $P = (-6, 1)$. Find the point obtained by:

a Reflecting P about the x-axis **b** Reflecting P about the y-axis

c Rotating P by 180° about the origin

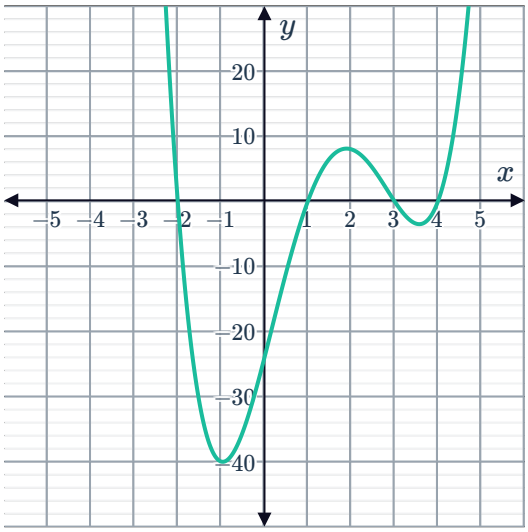
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22 Describe the symmetry of the graph of $x^2 + y^2 = 6$.

23 Suppose f is a function such that $f(3) = 2$. State a point that lies on the graph of f if:

- **a** The graph of $y = f(x)$ is symmetric with respect to the origin.
- **b** The graph of $y = f(x)$ is symmetric with respect to the y-axis.
- **c** The graph of $y = f(x)$ is symmetric with respect to the line $x = 6$.
- **d** f is an even function.
- **e** f is an odd function.

24 The graph of $y = P(x)$ is shown. Sketch the graph of $y = P(-x)$.



Dilations

25 Consider the function $f(x) = \sqrt{x}$. Write down the new function $g(x)$ which results from scaling $f(x)$ vertically by a factor of $\frac{1}{3}$, and scaling horizontally by a factor of $\frac{1}{2}$.

26 A function $f(x)$ is transformed into a new function $g(x) = f\left(\frac{x}{k}\right)$. If $0 < k < 1$, determine what will be the effect on the graph of $f(x)$.

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27 Consider the function $f(x) = \sqrt{x}$.

a Complete the table for $f(x)$:

b Sketch a graph of the function $f(x)$.

x	0	1	4	9	16
$f(x)$					
$g(x)$					

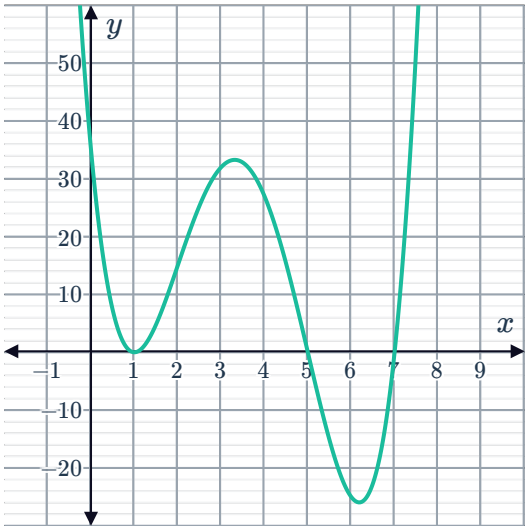
c Complete the table of values for the transformed function $g(x) = f(4x)$, giving all values in exact form.

d Sketch a graph of $g(x)$.

e Describe how $g(x)$ relates to the graph of $f(x)$.

f Explain why the transformed function $h(x) = f(-4x)$ can be undefined for the values of x in the table.

28 The graph of $y = P(x)$ is shown. Sketch the graph of $y = 2P(x)$.



Combinations of transformations

29 Suppose that the x-intercepts of the graph of $y = f(x)$ are -5 and 6. Find the x-intercepts of the graph of:

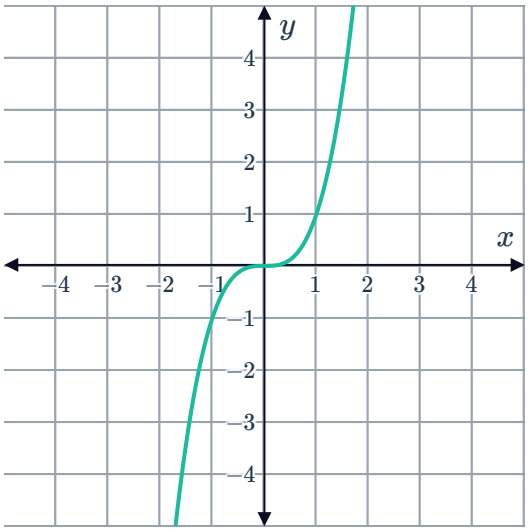
a $y = f(x + 4)$ **b** $y = f(x - 4)$ **c** $y = 3f(x)$ **d** $y = f(-x)$

30 Consider the function $f(x) = x^2 - 5$. Using function notation, describe the transformation of f that will result in the function:

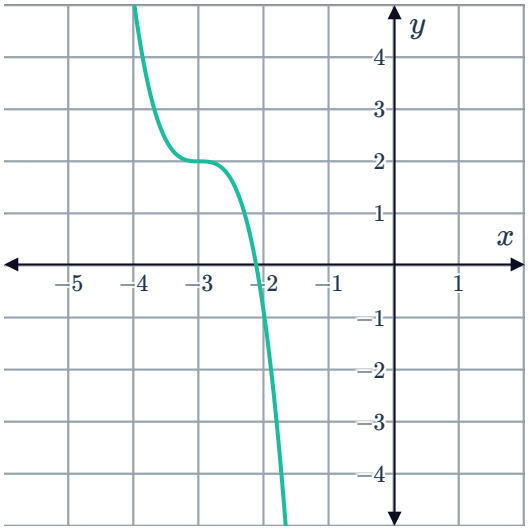
a $g(x) = x^2 - 6$ **b** $h(x) = 16x^2 - 5$ **c** $k(x) = 3x^2 - 15$

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31 Consider the graph of $y = x^3$:
Sketch the curve after it has undergone transformations resulting in the function $y = -4(x+4)^3$.



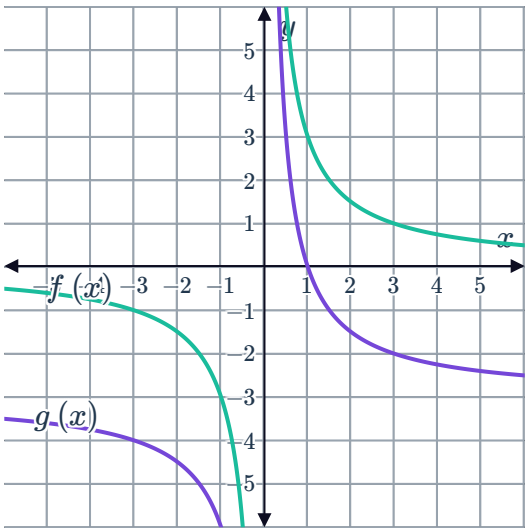
32 Consider the given graph:



a Describe the transformations of the graph of $y = x^3$ to the given graph.

b Write down the equation of the given graph.

33 Consider the graphs of $f(x) = \frac{3}{x}$ and $g(x)$ shown:

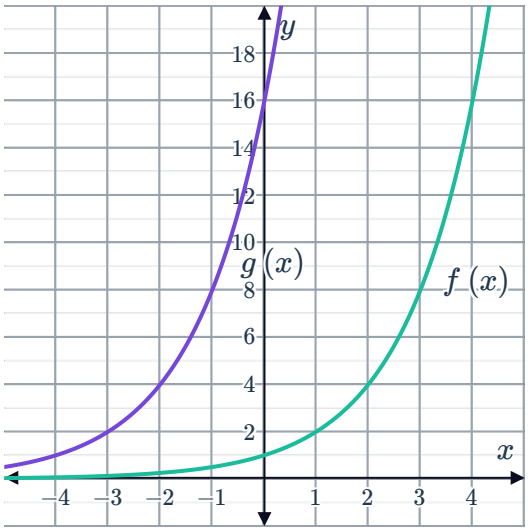


a Write $g(x)$ in terms of $f(x)$ using the transformation shown in the graph.

b State the equation of $g(x)$.

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34 Consider the graphs of $f(x) = 2^x$ and $g(x)$ shown.



a Write $g(x)$ in terms of $f(x)$ using the transformation shown in the graph.

b State the equation of $g(x)$.

35 For the following base functions, describe the transformations that have occurred:

a From $y = f(x)$ to $y = f\left(\frac{x}{4} + 3\right) - 5$

b From $y = g(x)$ to $y = \frac{g(x-3)+3}{2}$

c From $y = f(x)$ to $y = 2f\left(\frac{1}{2}x\right) - 3$

d From $y = h(x)$ to $y = 2(h(x) + 3)$

36 For the following functions, describe the transformations that have occured:

a From $f(x) = x^4$ to $g(x) = -7(x+4)^4$

b From $f(x) = x^4$ to $g(x) = -5x^4 + 8$

c From $f(x) = x^2$ to $g(x) = -10(x+8)^2 - 9$

37 The table below shows values that satisfy the function $f(x) = |x|$.

x	-3	-2	-1	0	1	2	3
$y = f(x)$	3	2	1	0	1	2	3
$g(x) = 5f(x)$							
$h(x) = -2f(x)$							

a Complete the table of values for each transformation of the function $f(x)$.

b Sketch the graph of $g(x)$.

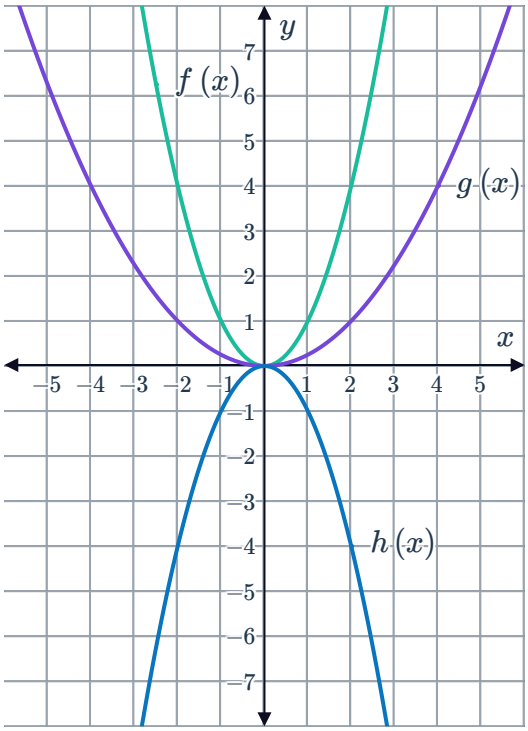
c Describe how to transform the graph of $f(x)$ into the graph of $g(x)$.

d Sketch the graph of $h(x)$.

e Describe how to transform the graph of $f(x)$ into the graph of $h(x)$.

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38 Three functions have been graphed on the number plane. $g(x)$ and $h(x)$ are both transformations of $f(x)$.



a State the equation of $f(x)$.

b Write $g(x)$ in terms of $f(x)$ using the transformation shown in the graph.

c State the equation of $g(x)$.

d Write $h(x)$ in terms of $f(x)$ using the transformation shown in the graph.

e State the equation of $h(x)$.

39 Some points on the graph of the function $y = f(x)$ are given in the table below:

Original point	$g(x)$	Corresponding point
$(9, -12)$	$g(x) = f(x) - 8$	
$(9, -12)$	$g(x) = 6f(x)$	
$(6, -7)$	$g(x) = f(x - 5)$	

Complete the table by finding the corresponding points on the graph of $y = g(x)$.

40 Suppose that $(-4, 3)$ is a point on the graph of $y = g(x)$. Find the corresponding point on the graph of:

a $y = g(x + 7) - 6$

b $y = -6g(x - 4) + 6$

c $y = g(6x + 1)$

41 The table below shows coordinates of points on the function $y = f(x)$. By performing the given transformation on them, find the corresponding transformed points.

Point	Transformation	Transformed point
$(-3, -1)$	$y = f(x - 5) - 2$	
$(0, 3)$	$y = \frac{1}{5}f(x - 3)$	
$(1, -2)$	$y = f(3x) - 2$	
$(-3, 5)$	$y = f\left(\frac{x}{2}\right) - 5$	

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